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Low Molecular Weight Supramolecular Gels as a Crystallization Matrix

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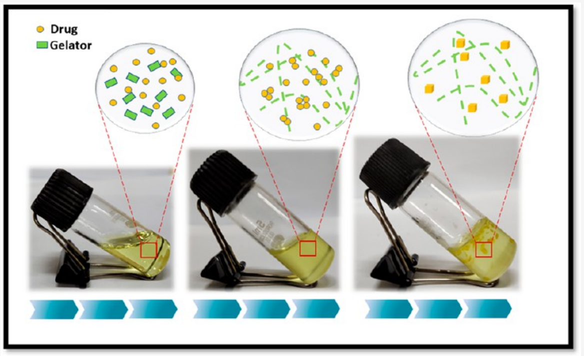

ABSTRACT: Advances in the solid-state formulations of active compounds have shown positive impressions on the drug's properties. Starting from the late appearance of the second form of the anti-HIV drug Ritonavir to the cocrystal-based drug Entresto has been a scientific effort at the crossroads of major subject areas like pharmaceutical development and crystal engineering. Most studies pertaining to the crystallization avenues have been the key choice to understand, predict, and control the solid form and properties, particularly relevant in the case of oral medicines. Crystallization from solution or slow evaporation is studied extensively and is a conventional rolling practice. However, dozens of other impending crystallization approaches encompassing vapor diffusion, precipitation, sublimation and melt crystallization, crystallization in the presence of additives, neat and solvent drop grinding, lyophilization, laser-induced crystallization, sonocrystallization, crystallization by ionic liquids and supercritical liquids, etc. are still being explored. The application of supramolecular gels as a crystallization matrix to nucleate the desired drug's phase preferably metastable forms has been under scientific investigation in the past decade. This Perspective discusses the strategic headway of using supramolecular organogels as a choice of crystallization media for small active molecules. The laboratory-shelved solvent trapped in the designed 3D network of gelator(s) acts as small confined containers to control the nucleation and growth process. Gel fibers act as nucleation-templating surfaces offering epitaxial growth and influencing crystallization outcomes via molecular recognitions and thus are highlighted. Besides highlighting the understanding of the physical properties of organogels, the techniques used to probe their properties and structures, and the thermodynamic concepts involved in gelator aggregation in organic liquids, this Perspective largely emphasizes the basic challenges that the drug manufacturing process faces related to the crystallization of the metastable phase and the role of a gel component(s) versus gel environment on the crystallization output.

1. INTRODUCTION

Heterogeneous nucleation is a common tool to counter the challenges related to the control of the crystallization output in traditional solution state processes. Crystallization is a complex event subject to a wide range of kinetic, thermodynamic, and molecular recognition parameters that proceed via nucleation and growth.[1−3] In contrast to homogeneous nucleation, primary heterogeneous nucleation involves nucleation onto solid substrates or surfaces of foreign substances.[4] The feasibility of the process lies in the decrement in the free energy barrier for the generation of the nuclei of critical radii to grow into a stable crystal. The subsequent growth is mostly a diffusion-dependent process of solute molecules depositing onto the surface of the nuclei.[5] The physical or chemical interaction of the solute with the heterogeneous surface results in the requirement of a smaller number of particles to form the nuclei of critical radii '*r*' compared to homogeneous nucleation.[6−8] The scientific literature has numerous models of heterogeneous nucleation like the Fletcher model,[9] classical flat surface model,[10] Turnbull's patch nucleation model,[10] and double spherical cap model,[11] to name a few.

The processing of pharmaceutical compounds in the solid state is the most sought after practice to achieve purity and reproducibility. Polymorphism in this domain is not always a favored phenomenon because of the safety and efficacy issues that are direct manifestations of the structure−function relationship.[12] The timeless example of Ritonavir marketed by Abbott Laboratories in 1996 in form I and its subsequent pull out of the market due to the appearance of form II leading to a failed dissolution test is a concern that still garners scientific and monetary investments. The metastable form I vanished from the entire scenario mysteriously, as the same manufacturing process could never reproduce it. Abbott Laboratories had to withdraw the capsules from the market

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Figure 1. Trend of publications from 2001 to 2022 accessed from the Web of Science with the term "supramolecular gel" or "low molecular weight organogels" in the search field of "Topic".

and reformulate them with Form II after intensive research over 2 years at the manufacturing units.[13] The curious case of this disappearing polymorph resurfaced with the reappearance of the metastable form I solely from common ball milling experiments as reported by Cruz-Cabeza and co-workers who stated that polymorphs with high energy cis conformers with respect to the carbamate group present in the drug molecule do become thermodynamically unfavored at the nanoscale dimension.[14] The gravity of such incidences in the pharma domain can be traced from the famous patent litigations that involve leading manufacturing companies standing against each other.[15,16] The increasing complexity of the molecular flexibility of modern active pharmaceutical ingredients (APIs) is critical to the development of polymorphic drugs.

Recent literature highlights the use of different techniques to control and screen polymorphic outcomes of crystalline compounds by means of seeding,[17] tailored additive crystallization,[18] nanoconfined crystallization,[19] substrate-directed crystallization,[20] sonocrystallization,[21] and laser-induced crystallization.[22] The heteronucleants used can be polymers, gels, small molecules, porous materials, etched glasses, and self-assembled monolayers (SAMs). In one such example, heterosurfaces were used to regenerate a disappearing polymorph form I of the compound 4-amino benzenesulfonic acid on treated heterosurfaces of 1,4-diaminonaphthalene seeds owing to the induced surface equilibrium.[23−25] The discovery of form VI of the drug sulphathiazole on SAM surfaces of mercaptobenzothiazole was reported by our lab.[26] However, the use of heterosubstances increases the possibility of compromising the chemical purity of the API. The possibility of the formation of eutectic, cocrystal, and complexes and the biocompatibility of the surfaces used are additional concerns.[27] The incorporated impurity can also lead to hurdles in gaining regulatory approval in due process. The ratio of the cost of manufacturing to market returns can never be ignored in the generation of SAM surfaces or porous materials to nucleate the desired form. Moreover, the feasibility of these processes is always in question during scaling up at the industrial benchmark of manufacturing, owing to the fluid dynamic interaction at large volumes and differential kinetics.[28] The systematic research on the use of gel as a medium to obtain single crystals arose from the appearance of Liesegang rings from potassium dichromate containing gel in the late 19th century. The advantage of utilizing gel as a crystallization media can be attributed to its ability to reduce the convection current of the media, control diffusion, and slow down the rate of nucleation to preferentially influence the kinetic and thermodynamic outcomes of the systems. The possibility to incorporate functionalized counterparts into gelators further enhances the epitaxial growth of the desired form of the compound.[29]

A web of science survey conducted with the specific terms "supramolecular gels" or "low molecular weight gels" or "molecular gels" in combination, in the search field "Topic" resulted in a staggering 71,393 publications over the period of 2001−2022. However, when the search is made more specific with the term "low molecular weight organogel" or "supramolecular gel" as the "Topic", it revealed an increasing trend over time in the related number of publications and citations received. From 41 publications in 2001 to 449 publications in 2022, it witnessed a nearly 10-fold increase in the amount of scientific literature generated in the domain per year (Figure 1). It indicates the tender age of the field which is yet to be explored to fill the voids in scientific understanding of the subject.

Abundant scientific literature is available for gels tailored to various applications; however, the content becomes limited and sparse when more specificity of low molecular weight organogels (LMOGs) is brought into perspective. The LMOGs are expected to have applications of the same order as polymeric gels in addition to the added utility due to their dynamic nature and responsiveness to different stimuli that open them up to easy chemical modification (Figure 2).[30−35] The potential of such materials in drug delivery is explored mostly when they can form hydrogels and are biocompatible with human physiology. However, for fat- and oil-soluble drugs, LMOGs are preferred for oral and parenteral formulations. The possibility of having a gel system of the active ingredient as a carrier could increase the efficiency in terms of sustained release, reduce drug loading, and be convenient for the known degradation and elimination profiles. A review published in 2021 efficiently lists some of the naturally obtained drug (paclitaxel, camptothecin, rhein, curcumin) based low molecular weight gels that promote slow drug release, over an extended period of time.[36]

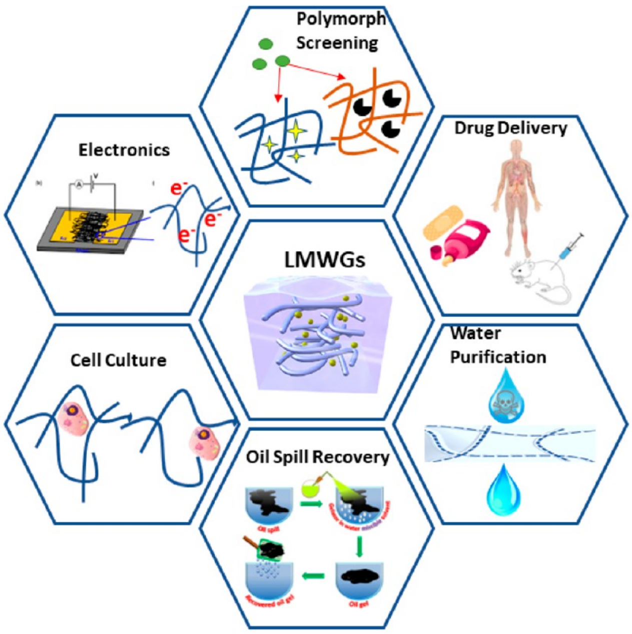

Figure 2. A few representative applications of low molecular weight gels (LMWGs).

Although much of the attention has been centered on hydrogels for their hydrophilicity, flexibility, and biocompatibility, concerns are encountered when considering their durability at subzero temperatures and fairly humid conditions. The advantages of incorporating organic solvent as a filling liquid into the gel increase the available options for a myriad of different solvents with specific characteristics, including the likes of mineral and vegetable oils as well. Another advantage of organogels is the scope of solvent tuning that can be achieved with a polar nonpolar mixture or polybasic mixture to attain an adjustable freezing point and boiling point of the solvent system.

Such improved parameters suitably increase the range of the working temperature for different applications involving the same. The most striking point of such a system is the applicability to compounds that are amphiphilic in nature, stabilizing the hydrophobicity as well as its hydrophilicity, standing up to the need of various research areas.[37] Reviews found in the literature mostly cater to hydrogels and their applications and in some cases to the organogels strictly focusing on the aspects of drug delivery. We present here the understanding and utilization of LMOGs as a crystallization matrix to probe nucleation and its effect on crystalline outputs for various drugs. This Perspective is broadly classified into four sections that highlight the design and syntheses of gelator molecules, the mesoscopic and macroscopic properties of the gel state and its subsequent characterization techniques, and most importantly the influence of the gel matrix on the crystallization of some solid crystalline compounds followed by factors influencing the overall process.

Gels are semisolid compounds composed of a network of fibers entrapping solvents in them. Depending on the solvent, entrapped gels can be classified into a hydrogel, organogel, and aerogel. However, depending upon the interactions or mechanism of forming a gel, it can be classified either as macromolecular or polymeric and supramolecular gels (Figure 3).[30] Unlike macromolecular gels, supramolecular gels are sustained solely by noncovalent interactions such as hydrogen bonding, $\pi \cdots \pi$ interactions, and van der Waals interactions forming the backbone of the growing fibers.[38]

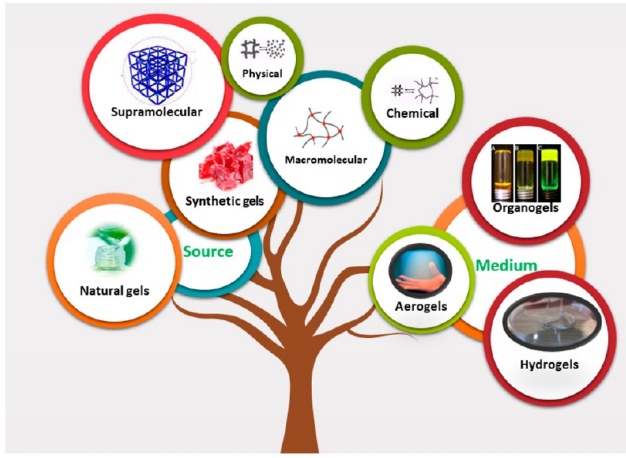

Figure 3. Classification of gels.

Although polymeric gels are known for their sturdiness, the thermoreversible nature of supramolecular gels somewhat adds leverage to their utility as a crystallization matrix. The dynamic, reversible, and stimuli-responsive nature of the self-assembly imparts tunable properties to the gels on the incorporation of different functional moieties into the gelator.[39] The term LMWG is a more specific term referring to supramolecular gelators with a molecular mass of less than 3000 Da. Diverse applications of LMWGs have been reported in the literature ranging from those in drug delivery, sensors, coatings, and culture media, to crystallization media some of which are tabulated in Table S1.[40] A simplified timeline of events from the first observed gel phase crystallization to the intentional application of LMOGs in a lab-scale pharmaceutical crystallization is presented in Figure 4.

2. DESIGN AND SYNTHESIS OF LMWGs

Although serendipitous, the first instance of a widely used LMOG dates to 1891. 1,3:2,4-Di-*O*-benzylidene-D-sorbitol and its numerous derivatives has a butterfly-like structure where the aromatic wings are available for functionalization by various acid and acyl hydrazide groups, resulting in compounds that can gel different solvents.[41] Another extensively studied LMOG is 12-hydroxystearic acid (12-HSA), which was studied to form gels back in the 19th century. During the 1930s and 1940s, the use of lithium salts of 12-HSA as lubricants in the aviation industry laid the foundation for modern lubricating greases. 12-HSA is a linear fatty acid molecule, where the presence of a secondary hydroxyl group not only infers chirality to the molecule but also increases the interaction probabilities with the solvent.[42] The strategic design has never been a universal one; it stems from a simple molecule and is later extrapolated to its derivatives obtained by various structural and stereochemical modifications. As discussed in the later sections, the idea of a gel state demands a balance of solvophobic and solvophilic interactions of the gelator with the solvent molecules, which is a function of temperature, concentration, and polarity. The self-complementary driving force for molecular aggregations of amides and ureas makes them a convenient choice to begin with apart from the library

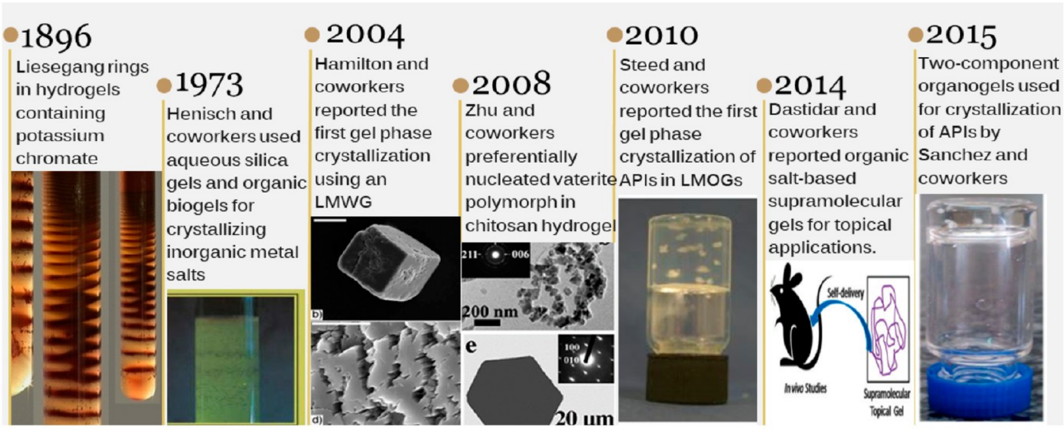

Figure 4. A timeline of the significant events toward crystallization mediated in gel media.

of molecules including carbohydrates, nucleobases, steroids, bile salts, and aliphatic hydrocarbons.[43]

Accordingly, the strategy of gelator design evolved from understanding the role of the different structural units of the reported gelators and advancement in the information on the structural insight into the gel state. A gelator consists of two counterparts: one responsible for aggregation to allow 1D fibrillar growth and the second to prevent it from crystallizing out of the solution, thereby facilitating the intertwining of 1D fibers to form the 3D network.[44] As a generalized strategy, hydrogen bonding functionalities and long alkyl chains are introduced in the target molecules to keep a check between the direction and disorderliness in the system. However, the presence of long alkyl chains is not hard and fast but rather a condition that prevents the 3D extension of crystalline domains. Some unconventional molecules, for example, dehydrobenzoannulenes (DBAs) based compounds, were also found to show gelation behavior for their one-dimensional π-stacking.[44] Hisaki et al. designed three boomerang-shaped DBA derivatives (Figure 5) based on dipole−dipole interaction and molecular symmetry.[45]

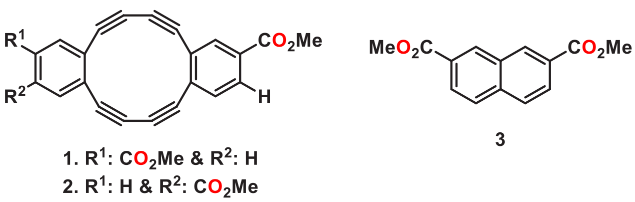

Figure 5. Dehydrobenzoannulenes (DBA) based gelators.

Despite the lack of long alkyl chains, **1** acts as an excellent LMWG for various organic solvents. Interestingly, when the methyl groups in the ester moiety in **1** were replaced by longer alkyl chains such as ethyl, propyl, and butyl groups, respective annulenes failed to show gelation (crystallize out from solution).[46] Thus, dipole−dipole interaction and molecular asymmetrization give anisotropically grown supramolecular fibers, which form a 3D network for gelation, whereas the presence of alternate π-stacking leads to crystallization. A class of two-component gel systems is the organic salts that form different supramolecular synthons to form 1D fibrillar networks. Dastidar et al. highlighted the application of crystal engineering principles to design gelators based on supramolecular synthons. The idea is to use synthons of primary ammonium monocarboxylate and secondary ammonium monocarboxylate that allow 1D growth for self-assembled fibrillar networks (SAFIN) (Figure 6).[47] Molecules with synthons that prefer 2D or 0D networks were not found to promote gelation.

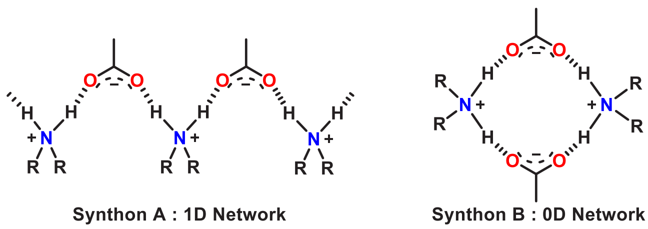

Figure 6. Schematic representation of two different kinds of hydrogen bond networks (HBN).[47]

The first gel phase crystallization using a supramolecular gelator was reported by Estroff et al. that involved a bis urea gelator flanked by an aliphatic core with hydrophilic functional groups at their peripheral ends.[48] The choice of solid materials to be crystallized in the gel media was a random one to the point where the crystallization of the substrate and formation of the gel were considered to be perfectly orthogonal. However, when shreds of evidence emerged on the influence of solid-state modification due to interactions between the high-energy gel surface and the substrate, the design of gelators became more substrate-specific. The idea of incorporating structural fragments into the gelator design to complement or mimic the substrate enhances the scope of better molecular recognition. To substantiate, the cis-platin mimicking gelator led to the discovery of the new dimethylacetamide hemisolvate of cisplatin.[49] Structural components may also be varied to improve the stimuli response of sol−gel transitions, enhancing the reversibility of the entire process. For example, in various photoreversible gels, the sol−gel transition occurs due to the functional group present in the gelator that responds to light of appropriate energy. A common strategy and design employed by Steed and co-workers to develop new gelators have a structural framework as depicted in Figure 7. The advantage of the design lies in the number of probable permutations and combinations that can be generated by altering the three basic units viz. core, linker, and peripheral units to build LMWGs with tunable properties.

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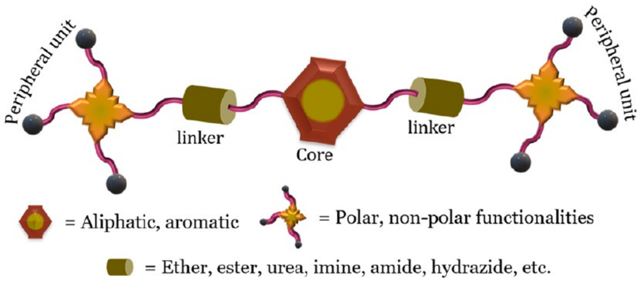

Figure 7. Schematic representation of a functionally modified gelator.

The process of gelation involves the dissolution of the gelators in an appropriate solvent by various stimulants, such as heat, light, ultrasound, pH, etc. (Figure 8). The hierarchical

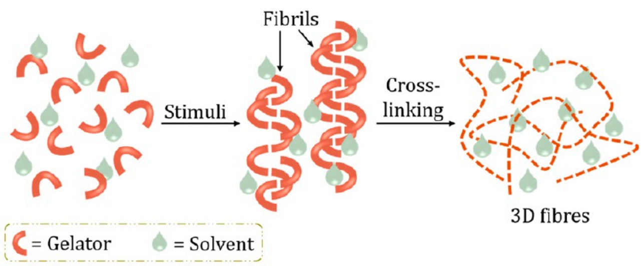

Figure 8. Schematic representation of the gel formation process.

self-assembly of the gelator molecules in the solution occurs via various noncovalent interactions followed by the formation of fibrils that collate to form bundles referred to as fibers eventually entangling in 3D space to immobilize gelling solvents by interfacial forces.[43] A gel state is a metastable phase, and the propensity to crystallize or precipitate out in the thermodynamic form depends on the subtle balance between the 1D and 3D interactions of the gelator molecules. Although gelation is often referred to as "crystallization went wrong" precisely it must be considered as crystallization that did not extend in perfect order in 3D.

For gelation, the gelator concentration must be equal to or greater than a particular value called the minimum gelator concentration (MGC), and the temperature must be equal to or lower than a particular threshold temperature called the transition temperature (T_{gel}) below which flow no longer is discernible over long periods. The possibility of synthesizing a gelator *in situ* in the reaction media that could gel the medium saves the process of screening solvents for gelation. For example, Suzuki et al. observed *in situ* organogelation by mixing highly reactive isocyanate (methyl 2,6-diisocyanatohexanoate and 2-isocyanatoethyl 2,6-diisocyanatohexanoate) and different alkyl amines.[50] They emphasized the shortening of the gelation time as gelators once formed can gel the mother solution (whereas in conventional methods, heating is required to dissolve the gelators) by self-assembling immediately into nanofibers, leading to rapid gelation.

3. GEL STATE: STRUCTURE, MORPHOLOGY, AND VISCOELASTIC NATURE

The difficulty in probing the gel state arises mainly from the variations in the self-assembly of the gelator molecules across many length scales. The investigation of the microscopic and mesoscopic properties of gels demands sophisticated instrumentation facilities. Insights into the gelator−solvent, gelator−gelator, and solvent−solvent interactions are necessary to quantify the thermodynamic and physical properties of the gel state.[51] The qualitative test, to begin with, is the vial inversion test where the flow of the formulated system subjected to its own weight is observed.[52] The MGC and T_{gel} values are firsthand indicative parameters of the stability of the gel. This method is highly prone to the inconsistencies generated by the type of container, the viscosity of the gelling solvent, and the thermal conditions associated. There is always a hierarchical trend in carrying out the characterization of the gel state with respect to various length scales (Figure 9).[53]

The formation of 1D self-assembly of the molecules by intermolecular interactions to generate gel fibers depends upon the nature of the gelator. The molecular packing at this level is often probed with the help of techniques like nuclear magnetic resonance (NMR) spectroscopy, infrared spectroscopy, diffraction techniques, and circular dichroism.[54,55] The chirality in a gelator has a significant influence on the gelation property in accordance with the "chiral bilayer effect".[56] In most of the cases, ^{1}H NMR proves that racemates are poor gelators in comparison with pure enantiomers. However, the reverse can also be true if there is a substantial preference for heterochiral packing among the homochiral fibers. A study of a chiral gel system of bis(amino)alcohol oxalamides with leucinol was observed to form inverse bilayers in lipophilic solvents for (S, S) and the racemate form of the gelator. The meso oxalamide form does not form any gel and crystallizes as monolayers. The racemate form in aromatic organic solvents is found to form stable gels consisting of enantiomeric bilayers. Temperature-dependent NMR data further established the presence of hydrogen bonding between the oxalamide and the OH groups and van der Waals interactions between *iso*-Bu groups of leucinol.[57] The diffraction studies of gels can reveal the spatial arrangement and directional nature of the gelators in the nanostructures forming the networks. Gels are often

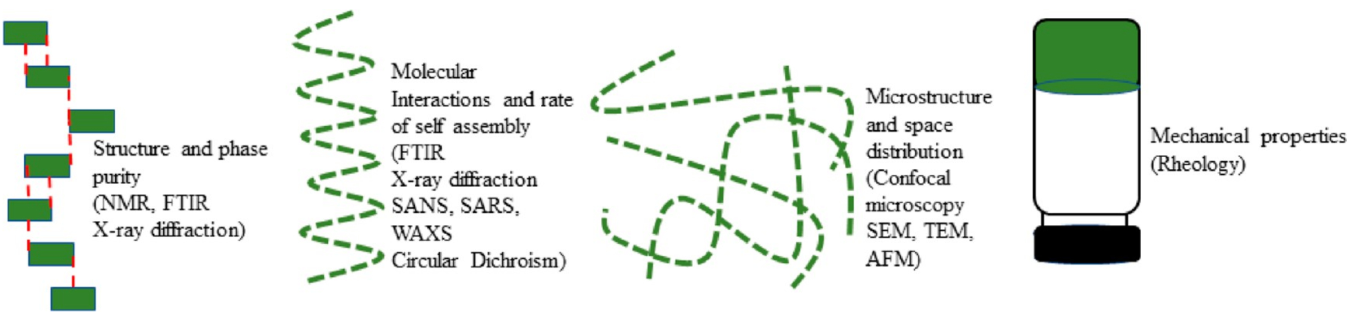

Figure 9. Hierarchy of gels across different length scales and associated characterization techniques.[53]

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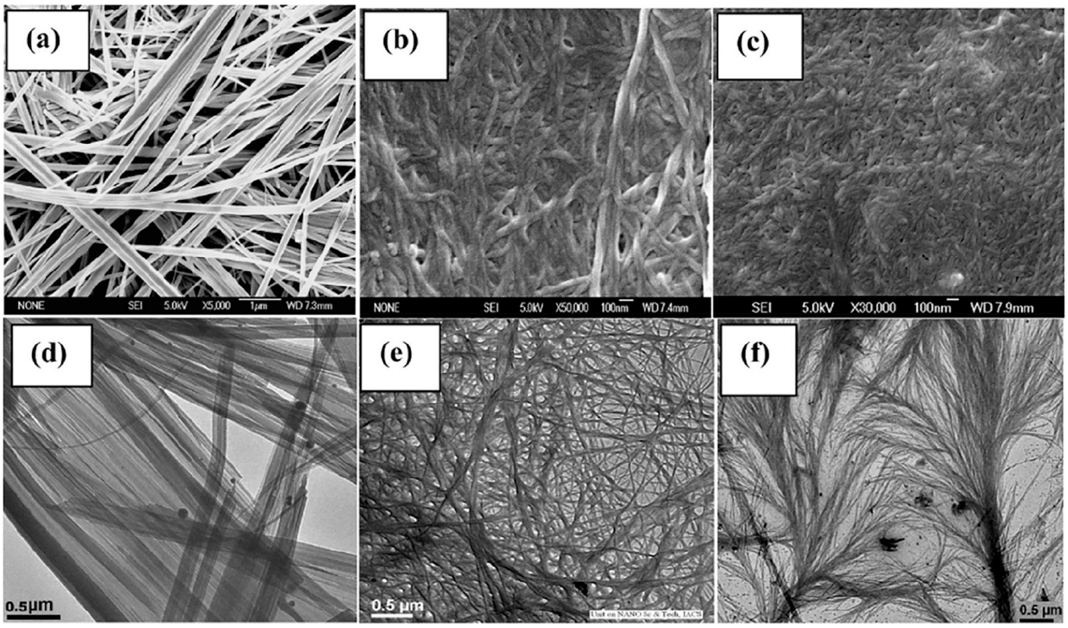


Figure 10. FESEM images of three dendritic gels (a), (b), and (c) and TEM images of the corresponding gels (d), (e), and (f).[60]

considered to be a state where the molecules are somewhere stuck between a crystalline and an amorphous state; hence, powder X-ray diffraction (PXRD) analyses generate distinct patterns corresponding to such crystalline domains. There are some disadvantages associated with the technique, owing to the interference introduced by the entrapped solvent molecules and crystalline impurities. Scattering experiments such as small-angle X-ray scattering (SARS) and small-angle neutron scattering (SANS), a technique similar to XRD at small angles, are used to study the skeletal organization of the molecules at the atomic level. The structural variations of 12-HSA during gel-to-sol transitions were investigated. The neutron scattering experiment established the presence of crystalline fibers that lose their density with increasing temperature without any effect on their cross-sectional area. The synchrotron X-ray scattering experiments revealed that gelation proceeded via nucleation and growth. The initial induction period had no change in intensity with respect to the time followed by the formation of an amorphous precursor, leading to the crystalline phase. The scattering profiles were reduced by (001) peak intensity indicating the crystallinity of the fibers.[58]

At the next hierarchical level, the entanglement, lateral thickening, aggregate morphology, and branching of the gel fibers at a microscopic dimension are investigated by techniques such as transmission electron microscopy (TEM), scanning electron microscopy (SEM), and atomic force microscopy (AFM). The field emission scanning electron microscopy (FESEM) or SEM imaging of gel systems reveals the 2D network structure with fibers emerging from different junctions and merging with each other. The SEM images of a few imide functionalized gel systems developed for the crystallization of barbital and thalidomide were observed to have helically twisted morphology in some cases and ribbon-like morphology in others.[59] A study where gels obtained from three different generations of dendrimers were subjected to FESEM analyses (Figure 10a−c) revealed a tapelike morphology for the first generation and twisted fibrillar morphology with an increased molecular weight of the dendrimer. However, TEM analyses of the diluted organogels presented a clearer picture of thinner fibrils that constitute the wider fibrils observed in the FESEM images (Figure 10d−f). AFM images also supported the presence of a nanofibrillar network in the gel system.[60]

Cryo TEM is a more preferable tool to remove solvent-drying artifacts during sample preparation. The microstructure of β-sitosterol:γ-oryzanol organogels in sunflower oil appeared as a uniform fibrous structure with individual fibrils of 10 nm diameter, indicating that gel formation involves not only self-assembly but also the aggregation of the fibrils by means of interfibril hydrogen bonds. The increase in junction point density with increased concentration proves the change in gelation behavior with the concentration of the gelator. AFM carried out at a higher resolution displayed the bifurcation and fusion of bundles into individual self-assembled fibrils running parallel-like ribbons unlike twisted strands mostly encountered in organogels (Figure 11).[61] Three-dimensional imaging of gels can be achieved by using confocal microscopy.

At a higher level of hierarchy, the mechanical properties of a gel system can be explored by rheology. Stress-sweep rheometry, which shows the storage modulus (G') to be at least 1 order of magnitude greater than the loss modulus (G''), characterizes the solidlike rheological features of supramolecular gels. Amplitude sweep measurements are used to

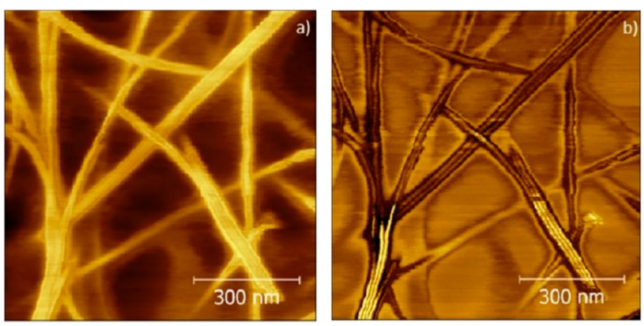


Figure 11. (a) Height and (b) phase AFM images of 10% w/w of β-sitosterol:γ-oryzanol gel.[61]

Table 1. Some of the LMWGs Used as Media for Crystallizing Drug Molecules since 2010

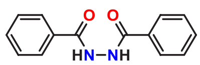
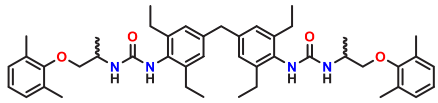
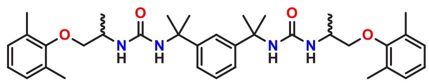
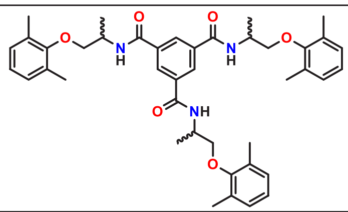
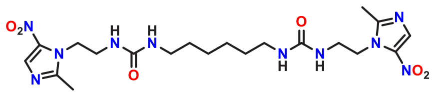
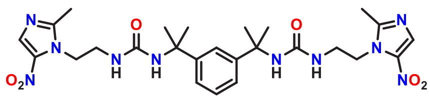
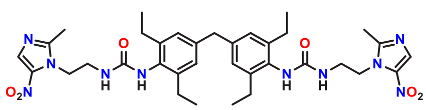
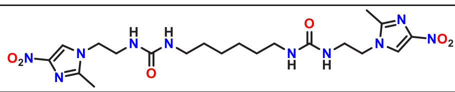
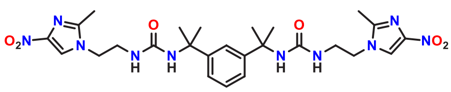
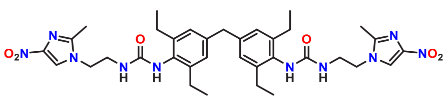
Gel No.	Low molecular weight gelators	Drug	Inference	Ref
4		theophylline, sulfamerazine, sulfathiazole, and flufenamic acid	Thermodynamically stable form III of flufenamic acid is nucleated with a change in crystal habit from block to needles.	75
5		mexiletine HCl	The gel of the compound in ethyl methyl ketone exclusively produces metastable form 3 of the drug	76
6		mexiletine HCl	The gel of the compound in 1,2-dichloromethane produces form 2 of the drug	76
7		mexiletine HCl	Produces the same phase of the drug as in solution crystallization	76
8		metronidazole	The gel in nitrobenzene was able to modify the habit of the drug from plate to needle shape	77
9		metronidazole	The gel in nitrobenzene was able to modify the habit of the drug from plate to needle shape	77
10		metronidazole	The gel in nitrobenzene was able to modify the habit of the drug from plate to needle shape	77
11		metronidazole	No habit modification was observed	77
12		metronidazole	Microcrystals of the drug were obtained	77
13		metronidazole	Thicker needle shape crystals of the drug were developed from nitrobenzene gel	77

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Table 1. continued

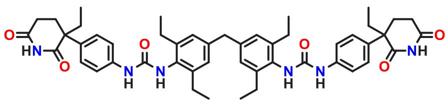
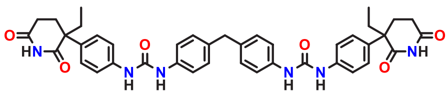
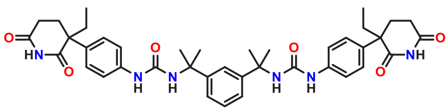
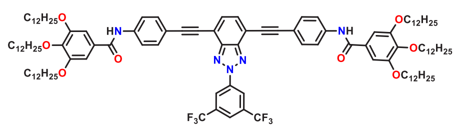
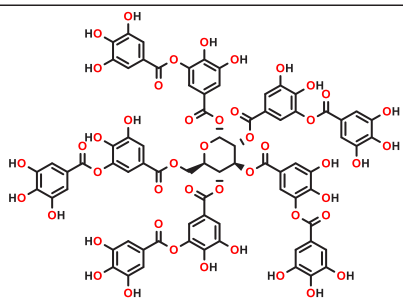
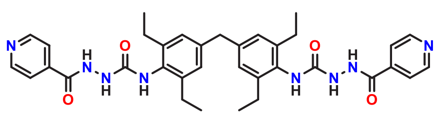
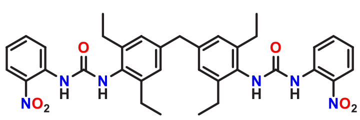
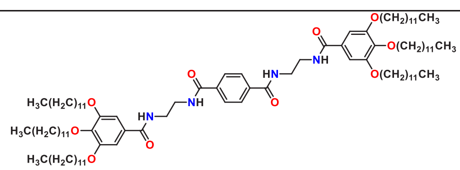
Gel No.	Low molecular weight gelators	Drug	Inference	Ref
14		barbital, thalidomide	The gel of the compound in few specific solvents could nucleate the metastable form III of the drug barbital. The nitromethane gel of the compound nucleates the kinetic form α of the drug thalidomide.	59
15		barbital	The gel of the compound in mixed solvent media could nucleate the metastable form III of the drug.	59
16		barbital	The gel of the compound in mixed solvent media could nucleate the metastable form III of the drug.	59
17		sulfathiazole, theophylline, sulfamerazine, niflumic acid	The methanol gel resulted in a habit change of Niflumic acid. The gel in 1-propanol resulted in the crystallization of Form I of sulfathiazole	78
18		caffeine, carbamazepine piroxicam	Size and morphology of caffeine crystals varied from that of control sample. Polymorph control was observed for carbamazepine and piroxicam in the gel media	79
19		isoniazid	Change in crystal habit of the drug in a gel media of the compound in propane-1-ol	80
20		ROY	The gel of the compound in toluene generates the metastable R form of the substrate molecule	81
21		aspirin, caffeine, indomethacin, carbamazepine	The gel in toluene produced the same forms as that obtained from solution crystallization.	82

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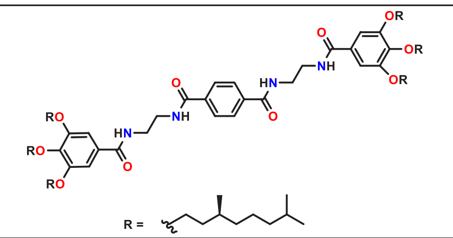
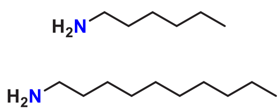
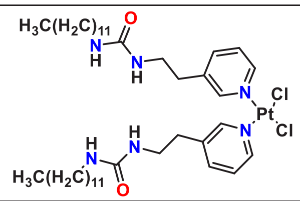
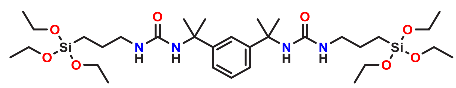
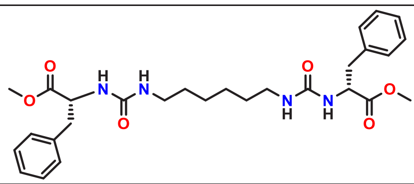
Table 1. continued

Gel No.	Low molecular weight gelators	Drug	Inference	Ref
22		aspirin, caffeine, indomethacin, carbamazepine	The gel in toluene led to concomitant crystallization of carbamazepine unlike that from solution.	82
23	LG2-Lys, DG2-Lys 	carbamazepine caffeine, aspirin, indomethacin	Aspirin and indomethacin did not crystallize whereas carbamazepine and caffeine did crystallize in the gel media.	83
24		cisplatin	Crystallization in the gel media resulted in the habit modification of DMF solvate and discovery of DMA hemisolvate.	49
25		carbamazepine	The gel in toluene could generate the metastable form II of the drug which transforms to form III within a day or two.	64
26		carbamazepine	High energy form I of the drug carbamazepine is crystallised from a gel of the compound in DMSO and water	64

determine the linear viscoelastic region (LVR), essentially followed by frequency sweep measurements to study its behavior in the LVR region.[62] Self-healing properties of a gel can be testified by using alternate step strain (or stress) measurements. In the case of a typical self-healing gel, G' is higher than G'' at low strain, G' and G'' are almost equal near the breaking strain, and G' dips below G'' at bigger stresses, but they revert to their former values when low strain is applied.[63] The rheological behavior of gel assemblies used for the purpose of crystallizing different analytes undergo various changes in their mechanical properties depending upon their binding affinity to the analyte in comparison to their tendency to self-associate. The roles of other variable factors such as solvent polarity and temperature is also found to influence the rheological data. In an example where carbamazepine (CBZ) that was loaded to a gelator in 1% (w/w) did not have any significant influence on the magnitude of loss and storage moduli, the yield stress however was found to increase by an order of magnitude. The apparent binding of the gelator to the drug (CBZ) as investigated by NMR titrations was found to be a bare minimum; hence, an increase in the yield stress was attributed to the physical support to the gel fibers provided by CBZ crystals. The yield stress is found to increase with the increase in gelator concentration.[64] However, the addition of substrate to the gel may not always lead to an increase in the yield stress; it may hamper the gelation as well, depending on the strength of the gelator-substrate interaction. Interestingly, in one of the reports, rheological behavior was found to be a function of the gelator to substrate ratio during crystallization of CBZ from a bis urea gel.[65]

4. CRYSTALLIZATION IN GEL MEDIA

Until the 1970s of the last century, the application of gel phase crystallization was limited to the inorganic materials in gels of silicate, gelatin, agar agar, etc. using a gel-diffusion process where reagents were added to form precipitates in gel media. Henisch et al. demonstrated that gel medium is beneficial in growing good-quality single crystals as it prevents turbulence, which permits the reagents to diffuse at a desirable controlled rate.[66] In addition, the gel provides a framework of nucleation sites that eventually hold the crystals delicately in the position of their formation, thus limiting effects due to the impact on the bottom or sides of the container (undue interaction with vial surfaces). Crystal nuclei formed in gels are well separated

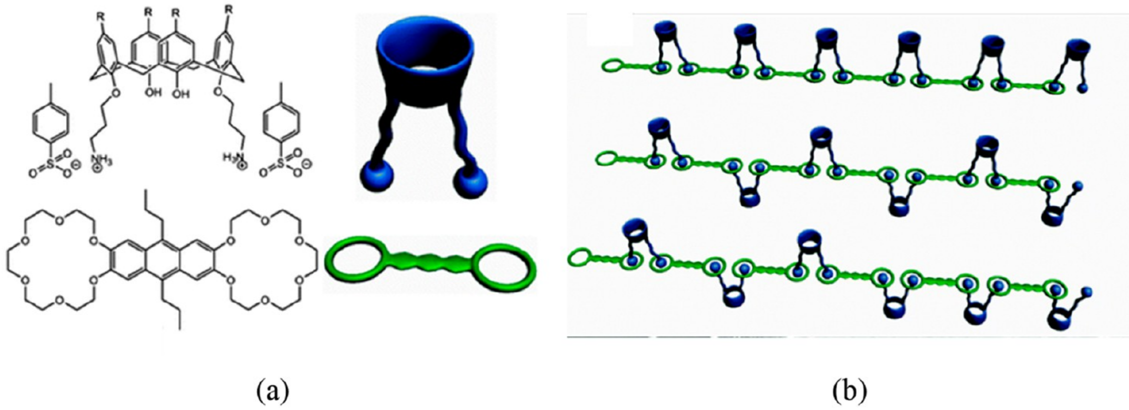


Figure 12. (a) Two components (co-gel) of calixarene tosylate salt and bis(crown ether) gel, (b) host−guest mechanism of gel formation in 1,2,4-trichlorobenzene.[84]

from each other, thus preventing precipitation. Moreover, the formation of a limited number of nuclei allows crystals to grow to a bigger size. Further, the manipulation of the growth environment can result in different crystal morphologies. The improved physical characteristics of gel-grown crystals over solution-grown ones attracted the attention of many chemists to use hydro/organic gels for growing crystals of organic molecules for structural elucidation through the single crystal X-ray diffraction technique.[67] The pores in a gel network act as small containers for crystallization, and the confined space of these pores controls the number, size, and rate of formation of the nuclei.[68] Moreover, the gel fiber surfaces can also influence the crystal outcomes by allowing its surface to act as a heterogeneous template for crystallization.[69] Zhu et al. reported the first preferential nucleation of the metastable form of calcium carbonate from a chitosan hydrogel.[70] The phenomenon of chiral amplification of a sodium chlorate conglomerate as observed in agarose gels can be highly beneficial for chiral switching of drug pharmaceuticals.[71] Functional groups in the gelator fibers can preferentially suppress the growth of a face in one direction to manipulate crystal morphology similar to directed crystallization induced by self-assembled monolayer (SAM) templates.[26,72] This may result from the strong interaction of the crystal face with the gel fibers, which makes it unavailable for subsequent growth. Morphological changes are very useful for improving the physical properties of the materials like flow rate, filtering/separation, packaging, tableting properties, etc. For example, Dastidar and his co-workers reported improvement in the flow rate of NaCl by habit modification from cubic to nearly globular rhombic dodecahedron-shaped NaCl crystals using glycine as a habit modifier.[73] Swift's group examined the effect of gel media on the habits of sodium bromate generated from crystallization in different agarose, gelatin, and silica gel media.[74] Growth from low-density agarose or silica gel resembles the growth conditions from aqueous solution, but from dense silica gels or at intermediate concentrations of gelatin a variety of habits like cubic, polyhedral, and tetragonal were observed. The advantages of crystallization in gel media are further enhanced by introducing low molecular weight gels to nucleate and control the solid-state behavior of drug pharmaceuticals where the importance is significantly realized.

4.1. LMWGs in Pharmaceutical Crystallization. The use of a low molecular weight gel matrix for the crystallization process has the potential to address various solid-state issues for pharmaceuticals of concern. The crystallization processes are designed to achieve control over shape, size, and homogeneity in crystalline phases, control of concomitant crystallization, polymorph screening and discovery, separation, and enantiopurification. Table 1 represents a summary of the available literature in the domain of supramolecular gel phase crystallization involving different drug molecules and the inferences deduced from them.

4.1.1. Discovery of New Solid Forms. Gel phase crystallization can result in the discovery of a new polymorph or solid form of a drug molecule provided the interactions at the molecular level favor the same. Although it is difficult to predict or design a gelator for this specific purpose, the idea is to incorporate a substrate mimicking functionality into the gelator or to use molecular recognition chemistry. The development of gelator **24** utilized for the crystallization of cisplatin resulted in the discovery of the second known solvate form of cisplatin incorporating one *N*,*N*-dimethylacetamide (DMA) molecule per two cisplatin units. The crystallization was carried out using a bilayer gel−sol technique where a DMA solution of cisplatin was injected into the gel of **24** in *o*-xylene. The crystallization carried out in a few other related gel media also resulted in the formation of DMA solvate of cisplatin along with α- and β-cisplatin. The control experiments in DMA solvent initially did not produce the solvate; however, after a few months of the gel phase experiments, the control experiment did show the presence of DMA solvent, which is reported to be due to unintentional seeding of the new solid form. The discovery of the new solid form was not attributed to the presence of the mimicking part, but its influence on the crystallization was accounted for with critical observations. The probable interaction between the gel and the substrate molecule via Pt−Pt stacking and amine-chloride hydrogen bonds was considered to be the reason that encouraged the formation of a new solid form.[49] Another well-known example of polymorph discovery is that of chlorphenesin, an antifungal agent that was crystallized from a two-component calixarene-based gel in 1,2,4-trichlorobenzene (Figure 12). This calixarene-based two-component gel uses the principle of host−guest chemistry and serves as an efficient medium for the crystallization of hydrophobic drugs.

The discovery of a new polymorph or solid form requires defined control over the entire process of nucleation, which is also dependent on other intrinsic factors like temperature, pressure, humidity, polarity of the solvent, etc. Although it is too early to comment, a gel phase may be considered to provide the necessary intrinsic environments better than what a solvent alone can, to nucleate novel solid forms.[84]

4.1.2. Selectivity for Solid Forms. Most of the gel phase crystallization outcomes of pharmaceutical compounds were seen to have bias or selectivity toward a particular solid form. The underlying mechanism for polymorph control lies in manipulation of the degree of interaction between the drug and the gel. The expected aim of these crystallization experiments is to stabilize the metastable kinetic forms of the drug, which are often found to have better physicochemical profiles. In 2010 Steed and co-workers for the first time reported the LMW gel phase crystallization of CBZ using gelators **25** and **26** in toluene and different mixed solvents. CBZ is a highly polymorphic drug with four known polymorphs and one hydrate whose crystallization behavior is highly affected by the rate of cooling and agitation. The controlled experiments in pure solvent resulted in a mixture of form II, form III, and the dihydrate. Interestingly, the crystallization from gel **25** in toluene and a chloroform:toluene mixture resulted in the generation of kinetic form II exclusively, which later transformed into the thermodynamic form III. However, the transformation of the forms was slowed down in gel media as compared to that in pure solvents. The slowed kinetics of dissolution of kinetic form II was reported to be the rationale behind this observation. Additionally, the appearance of high energy form I from the gel of **26** in DMSO:water was different from what was obtained from the control. Form I was reported to have been produced from melt above 170 °C and at relatively low humid conditions prior to this gel phase crystallization report.[64]

The selectivity toward solid forms can address the issue of concomitant polymorphism, a common phenomenon in the crystallization of numerous APIs. Drug memetic gelators **14**–**16** containing an imide functional group were used to study the effect of gel media on the concomitant crystallization behavior of barbital (BAR) and thalidomide (Figure 13). To find out the effective role of the drug mimicking the functionality in the gelators, crystallization was also carried out in solution and nonmimetic gels. Control over the concomitant crystallization, high selectivity toward the kinetic form of the drugs, and failure of nonmimetic gelators to prevent concomitant crystallizations indicated that the viscous gel media is not primarily responsible for the outcomes; instead, the role of drug-mimetic gelators was necessary to prevent the concomitant crystallization of the drugs.[59]

Mimicking a conformational aspect of a drug molecule in the gelator was also found to lead to polymorphic control in the case of ROY (Red Orange Yellow). ROY is a precursor to the drug olanzapine and exhibits a high degree of polymorphism, with 11 experimentally recorded polymorphs. The bis(urea) gelator **20** mimicking the chemical structure of ROY was reported to crystallize the metastable form (R form) at an optimal drug loading in the gel media contrary to crystallization from solution and other nonmemetic gels that resulted in the thermodynamic form (Y form). They hypothesized that incorporating molecular features into a gelator that mimics ROY would increase the probability of influencing crystal growth. They found that the lowest energy calculated structures (with extended conformers) from crystal structure prediction methods possessed considerable similarity to the experimental xerogel XPRD data and proposed them as a model for the compound. In the proposed model of extended

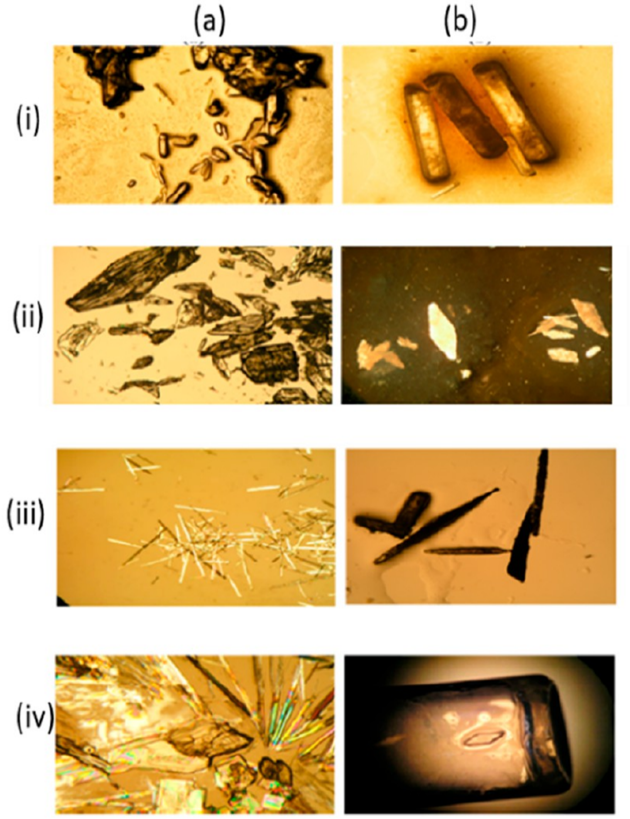


Figure 13. Microscopic images of BAR crystals produced from (a) solution crystallization and (b) **14** gel phase crystallization in (i) 1-butanol, (ii) ethanol, (iii) nitromethane, and (iv) cyclohexanone, respectively.[59]

conformers, nitroaniline is exposed on the surface of gel fibers available to interact with ROY molecules. The proposed conformer of the gelator gives a better steric and electrostatic match to the R than the Y conformation of ROY explaining the preferential growth of the R form of ROY in the gel phase crystallization. These results suggest how the gel fibril's surfaces can manipulate the crystallization outcomes by gelator-solute interactions.[81]

The reverse is also observed where the gel media results in a concomitant crystallization, contrary to a solution that produces a selective form of the solid substrate. Sanchez's group reported amplification of chirality of linear tetra-amide LMWG (achiral **21**, and chiral **22**) and used the respective gels in the crystallization of drugs (namely, aspirin, caffeine, indomethacin, and carbamazepine). The gel phase and solution phase crystallization generated the same polymorphic crystal outputs of the APIs except for CBZ. CBZ crystallized in the organogel prepared from **21** and **22** at 1 wt % and mixer of **21** and **22** in a 9/1 ratio resulted in polymorph III (in the case of **21**) concomitant mixture of polymorphs II and III (in the case of **22** and the mixer of **21** and **22**), whereas solution crystallization of the control experiment produced polymorph III.[82] The changes observed in the polymorphism of CBZ were ascribed to the retardation of diffusion, nucleation, and convection currents present in the crystal growth media and/or to a heteronucleation phenomenon. Similar findings were also observed when they discussed the crystallization of CBZ from two component gels of **23**, leading to concomitant crystallization of form II and III unlike what was obtained from

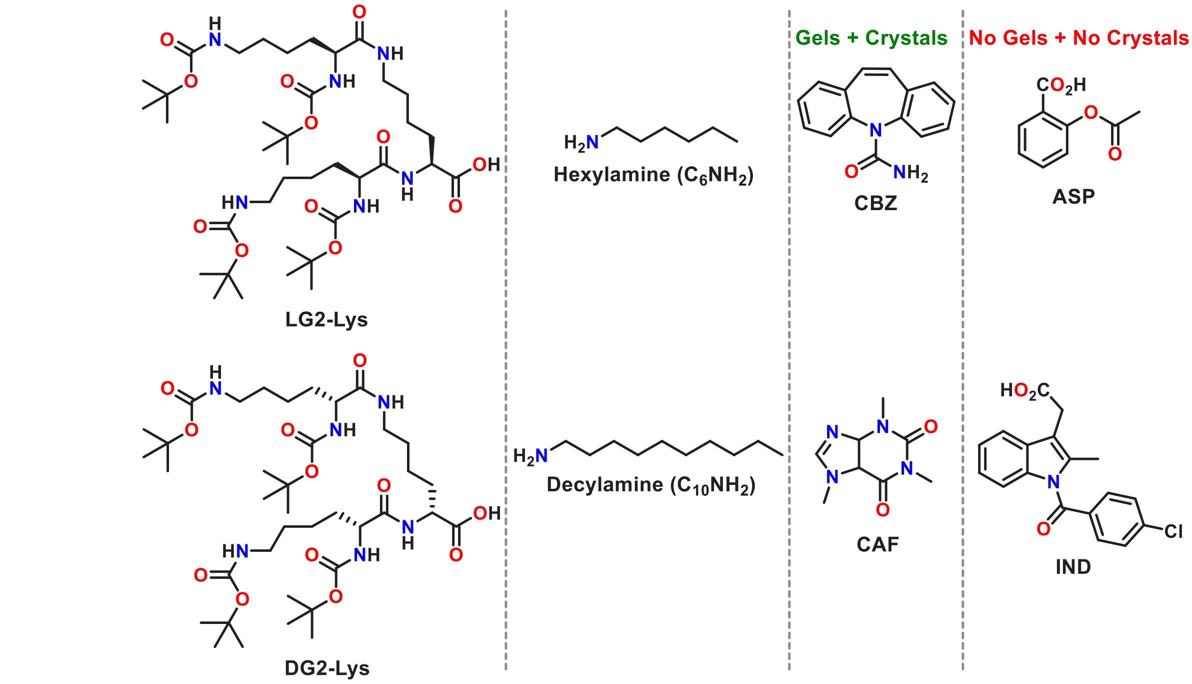


Figure 14. Schematic representation of the synergistic and antagonist effect of drug molecules on gelation.[83]

solution. Further, useful insight was drawn into the synergistic and antagonistic effects of the functionalities of drugs on gelation and crystallization. They synthesized two different dendrons LG2-Lys and DG2-Lys with different functional groups and used two different primary amines of varying chain length to prepare 4 different gels in combination (Figure 14). Drugs such as caffeine (CAF) and CBZ with no acid functionality do not directly interact with either of the two components and readily crystallize inside the gel network. APIs with a carboxylic acid group, aspirin (ASP), and indomethacin (IND) inhibited the gelation and crystallization of the drug. This behavior is the result of the relative binding constant for different competitive interactions of a "$-CO_2H\cdots NH_2-$" pair present in the crystallization system. Thus, it is important to analyze the competitive interactions among gelator/drug/solvent to develop a robust system to control crystallization in the gel phase.[83]

The selectivity toward a kinetic form is not a universal case. Very often the thermodynamic form of the drug is encountered in the gel media for obvious reasons of slower nucleation and growth kinetics. Gelator **4** used for the crystallization of flufenamic acid resulted in the crystallization of the thermodynamic form III of the drug compared to form IV obtained from solution.[75]

4.1.3. Habit Modification of Solid Forms. As the gelator fibers are potential templates for nucleation, the functional group(s) present in the gelator may interact with the crystallizing substance and can influence the morphology by controlling the growth of the crystal in a particular face(s). For example, gel-phase crystallization of metronidazole (a nitroimidazole antibiotic) showed marked habit modification of the crystals in gels compared with crystallization from nitrobenzene solution (Figure 15). The gelators **8–10** had metronidazole, and **11–14** had its isomer isometronidazole as peripheral functionality to mimic the crystallizing substrate.

The urea motif in the gelator was expected to form α-tapes in fibrils that could align the drug-derived functional groups as

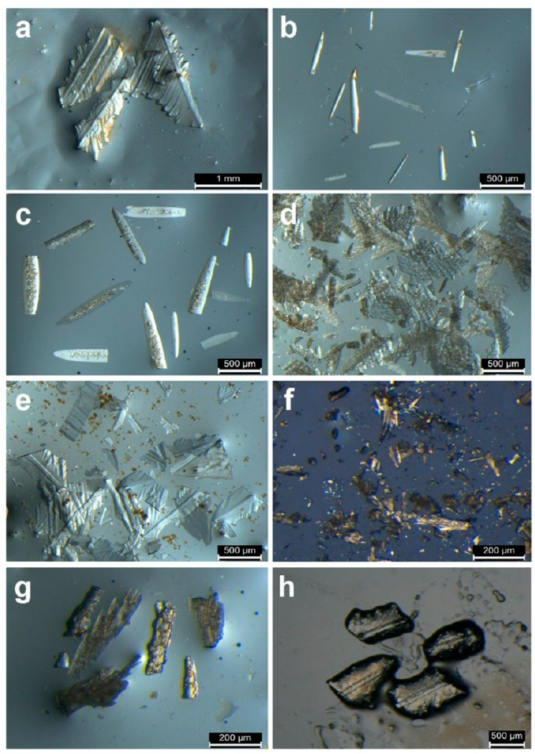


Figure 15. Comparison of crystals of metronidazole from solution and gel phase crystallization in nitrobenzene. (a) Solution phase and (b) gel phase crystallization in **8**, (c) **9**, (d) **10**, (e) **11**, (f) **12**, (g) **13**, and (h) nonmemetic gelator, respectively.[77]

a locally ordered array on the surface of the gel fibers. The gel phase crystallization with different mimetic gels resulted in different morphologies: for isometronidazole based gels (**11–13**); the morphologies were found to be plate-shaped (**11**), microcrystals (**12**) and thicker needles (**13**) whereas in metronidazole based gels (**8–10**) needle-shaped crystals were observed. The needle-shaped crystals obtained from gel **10** were seen to be clumped together in a herringbone fashion. In contrast, solution phase crystallization resulted only in plate-shaped crystals aggregated in a herringbone fashion. Face indexing of crystals grown from solution and gels revealed a complex and varied influence of all of the gels on morphology. Metronidazole crystals from solution have faces mainly with low indices, like ($\bar{1}00$), ($0\bar{1}0$), and ($00\bar{1}$), as well as a large ($1\bar{1}5$) face. The crystals grown from gel expressed some higher index faces, especially those grown from gel **10** exhibited a wedge shape with faces like ($0\bar{9}7$), ($3\bar{5}7$) of the order (Figure 16).[77]

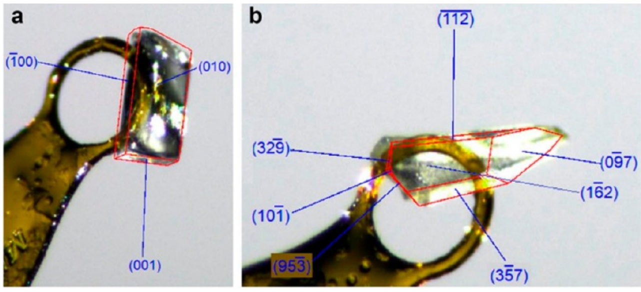


Figure 16. Face-indexed morphology of metronidazole crystals (a) grown from nitrobenzene solution, (b) wedge-shaped crystal from a nitrobenzene gel of metronidazole-mimetic gelator **10** expressing higher-order faces.[77]

These differences in crystal habit modification were attributed to the adsorption of the growing crystals onto the surface of the gel fibers leading to molecular epitaxy for the mimetic gels. In another such example, a supramolecular metallogel of tannic acid was used for the crystallization of three different APIs. They reported various critical observations on the size and morphology of the crystals grown in the gel media. The crystallization of caffeine (CAF) in gel media was observed to be influenced by the concentration of CAF loaded to the gel, temperature, and TA:Ti(IV) stoichiometry. With the increase in TA:Ti(IV) stoichiometry, uniform needle-shaped crystals of CAF changed to hairlike structures on rodlike cores and eventually to leaf-like structures. A better influence on the shape of CBZ crystals was observed with the change from needle to spherical morphology in control experiments and gel phase experiments, respectively.[79] The change in habit and morphology is a common observation of most of the gel phase experiments. As reported in Table 1 the observations support the same fact. However, computational studies on the probable drug:gel fiber interactions correlating their manifestation in the exhibited morphology are sparse.

The limited scientific data discussed above in these three different sections indicate the potential of gel media to address concerns of various solid forms and understand heterogeneous nucleation better. We can infer that protruding functionalities and conformational aspects of the gel fibers have a conclusive impact on the crystallization of various substrates. The strength and directionality of these interactions at the gel substrate interface determine the fate of the process. Relatively stronger interactions between gelators and the substrate molecule may be antagonistic to the entire process of gelation. An appropriate density of these interactions can sustain gelation and provide sites at the fibrillar interfaces for directed nucleation to induce polymorph selectivity. The moderate to weak interactions at the fiber substrate interface mostly affect the shape and size of the crystalline outputs by prohibiting growth along certain crystalline faces and enhancing the same for others.

5. ROLE OF VARIABLES

Starting from the gelators' structure, gelling properties (scope of solvent, gelator concentration), including thermal and mechanical strength, drug loading, temperature, solvents, time, and various other intrinsic factors, play a crucial role in affecting the outcome of gel phase crystallization. The process may not proceed as expected or planned due to hindrance in gelation by the substrate, complexation, or cocrystal formation by strong preferential interactions and difficulties in recovery of the crystalline material from the gel media. Some of the important variables that affect the gelation and further the process of crystallization are discussed in the following sections.

5.1. Solvents.

It is believed that gelators mutually competing for gelation and crystallization should have intermediate solvophilicity/solvophobicity. The role of solvents in gel formation is interpreted by considering solubility and thermodynamic parameters of solvent-gelator interactions.[85,86] The interaction among gelators or the interaction between gelators and solvents can induce the dispersed particles to connect and form a 3D network structure, rather than gathering to form big particles and settle down. The contribution of the solvent to the gelation behavior can be interpreted in terms of variation in MGC, the solvent−gelator specific (i.e., H-bonding), and nonspecific (dipole−dipole, dipole-induced, and instantaneous dipole-induced forces) intermolecular interactions, polarity, and viscosity of the solvent. Solvent properties influence the solubility of molecular gelators, which mediate the aggregation process of gel formation and other types of aggregation like precipitate, crystal, fibers, and other structures.

The influence of the specific and nonspecific forces on gelation is critical for understanding the process of gel formation. In particular, the correlation of gelation number (no. of solvent molecules per gelator molecule) to solvent parameters can provide useful information on the specific interactions that dominate gel formation. Among various empirical solvent scales, the Hildebrand solubility parameter[87] (denoted as ä) is derived from the cohesive energy density of the solvent and directly reflects the total forces that hold solvent molecules together.

In terms of the thermodynamic viewpoint, gelator−gelator and gelator−solvent interaction energy (enthalpy of dissociation, solvation, etc.) can be seen as the driving force for the gelled state. The Hansen solubility parameters (HSP)[88] approach can be a powerful tool to expand the scope of solvents that form gels, which should ultimately increase the utility of each gelator. It describes the cohesive energy density of the solvent using three contributions: hydrogen bonding interactions (δh), van der Waals or dispersive interactions (δd), and dipole−dipole or polar interactions (δp) (Figure 17). The Hansen space of each gelator can be used to form 3D Hansen plots by fitting a large data set containing solvents that both promote and disrupt gelation to identify additional solvents for gel formation. Other available methods like Kamlet−Taft parameters,[89] Flory−Huggins parameter,[90] etc. are also useful for understanding the role of solvent. The HSP sphere of a gelator allows us to predict the probable gelation of

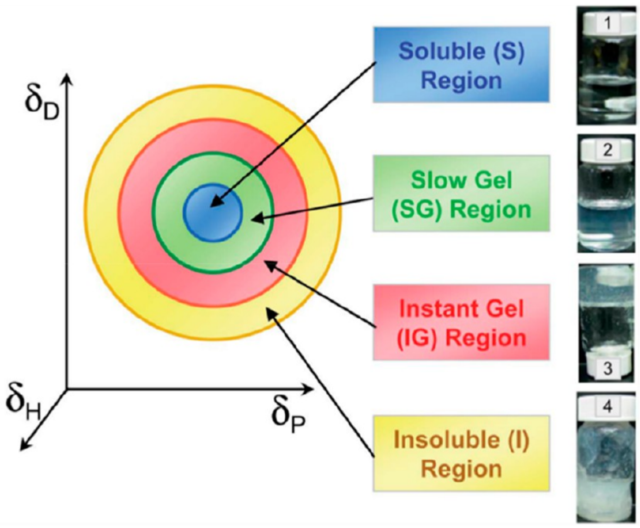


Figure 17. Representative 3D plot of Hansen solubility: plotted for the various solvents where the axes represent the three Hansen solubility parameters (δ_D = dispersive, δ_P = polar, and δ_H = hydrogen-bonding interactions).[88]

a particular solvent. It also helps to choose a combination of solvents for gelation based on the calculated parameters for different solvents.[91]

H-bonding has been shown to play an important role in the self-assembly of most LMWGs. Solvents can interfere with H-bonding self-assembly of gelators and can promote or interfere with the gelation process. An interesting observation by Vidyasagar et al. found that cyclohexane-1a,3a-diol derivative gelators underwent crystallization at low concentrations and gelation above the MGC in the same solvent (Figure 18).[92] However, a proper study revealed that the amount of water in the solvent decided the fate of crystallization or gelation. Thermogravimetric analysis (TGA) of the crystal and xerogel confirmed that the crystal contains water molecules. Thus, water present in the solvent interacts with the gelator, preventing it from forming a gel.

Braga et al. reported a gelator of silver(I) complex of 1-phenyl-3-(quinolin-5-yl), which forms gel or crystals in the same gelling solvent depending on the process.[93] When the gels were allowed to dry in open vials, xenogels were obtained. The gels were allowed to dry in sealed vials, resulting in crystals. The observations made were not solely due to the solvent behavior but rather a prominent effect of the kinetics of evaporation. However, crystallization of the gelator from different solvents yielded different polymorphs such as CH_3CN resulting in Form I, EtOH in Form II, i-PrOH in Form III, and MeOH in Form IV (Figure 19).

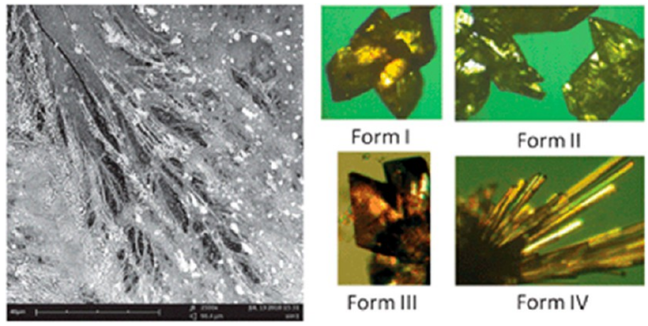


Figure 19. SEM image of the xerogels obtained by drying in the air the 1:2 $AgNO_3$:PQ5U gel from EtOH (left) and polymorphs crystallized directly from its gels (right).[93]

The selection of solvent(s) is important for gel phase crystallization as the solubilities of both the gelators and the drug must not be at extreme opposites in a given solvent. The crystallization of APIs in LMW gels is a solvent-specific phenomenon. The polymorphic control or selectivity in the gel media is seen to vary from solvent to solvent.

5.2. Concentration. The gelator concentration is crucial for gel formation. As discussed above, no gelation will be observed below MCG. The increase in gelator concentration mostly leads to improvement in gel properties as observed from various quantitative parameters (e.g., T_{gel}, gel fiber density, rheology, etc.). The time of gelation is reduced by an increase in concentration because of the reduction in gelation

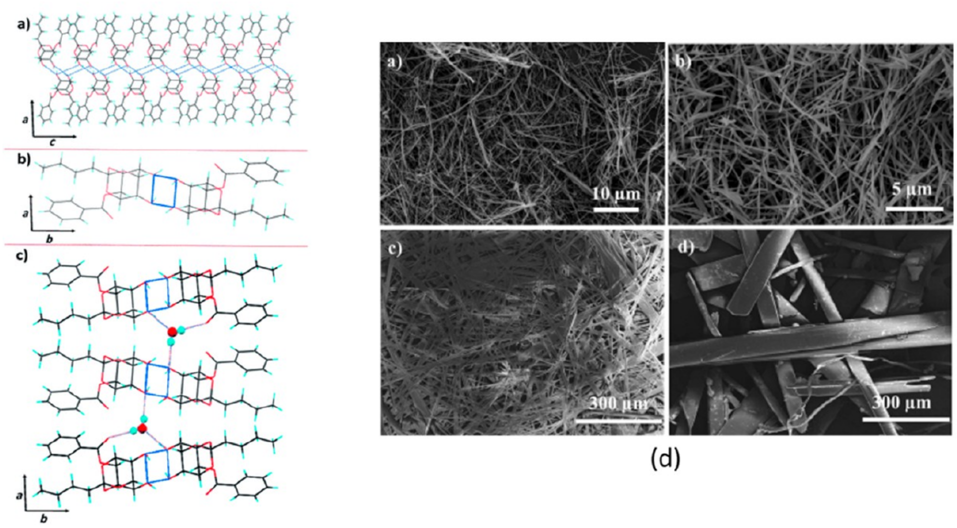


Figure 18. Zig-zag arrangement of gelator molecules forming a fibril: (a) viewed along the *b* direction; (b) viewed along the *c* direction; and (c) water molecules connecting adjacent fibrils along the direction to form a 3D crystal; (d) SEM images of xerogels and crystals.[92]

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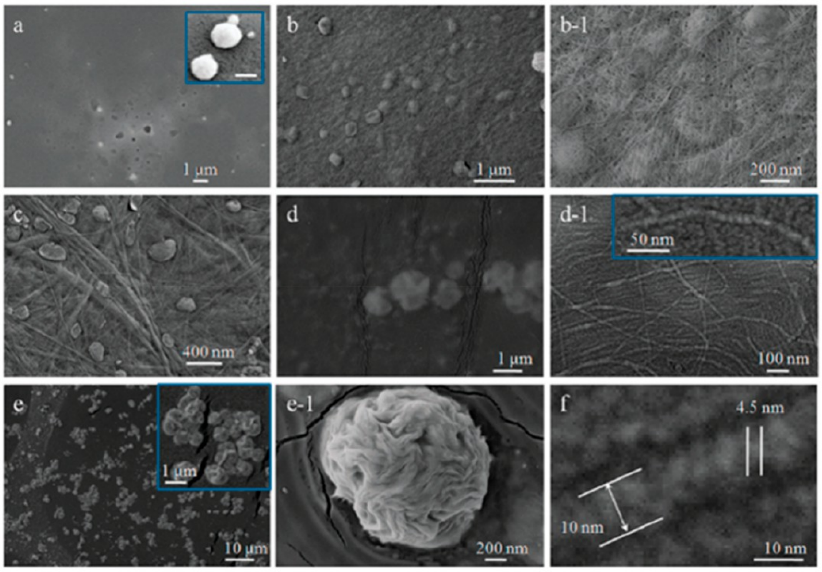

Figure 20. Morphological evolution vs folic acid concentration of gel. SEM images, (a to e): samples with concentrations of 1 mM, 5 mM, 10 mM, 20 mM, 30 mM (v/v, 6/4) and b-1, d-1, e-1: enlarged images of (b), (d), and (e) respectively, (f) amplified images from images of 5 mM; insets: magnified images of vesicles (scale bar: 100 nm), twisted fibers and nanoparticles.[94]

number. Folic acid is a well-known gelator that forms gels in different solvent ratios of the water/DMSO mixture. The gel morphology at different concentrations is observed from the SEM images of the gel. Concentration played a major role, as it allowed the gelators to grow different assemblies like fibers, vesicles, nanoparticles, etc. Depending on the nature of these assemblies, their properties vary (Figure 20).[94]

The concentration of the drug loading is also an important aspect of the crystallization of APIs in the gel media. The selectivity for the high energy form I of CBZ as reported from the gel of **26** could only be obtained after the drug loading had crossed a minimum threshold.[64] The optimum drug loading must be evaluated well for the process to work efficiently. As both crystallization and gelation are affected by self-assembly and supersaturation, the outcomes of gel phase crystallization mostly depend on the strength of the gelator-solute interactions. Generally, these two processes are orthogonal where the gel is formed first, and then in that gel phase nucleation and crystal growth of solute occur.[69] In certain cases, both events occur simultaneously in a competitive manner to influence the outcomes. In this aspect, Dawn et al. investigated the effect of supramolecular gel phase crystallization of carbamazepine (CBZ) in a bis(urea) based gelator. They used small-angle neutron scattering (SANS) and other tools to observe the influence of CBZ in the structural evolution of the gel. SANS data suggested that the CBZ suppresses fiber entanglement due to the competitive nucleation events involving gelation and crystallization. This competition climbs its peak at the intermediate concentration of gel (0.12%) and is found to be driven by the concentration and the relative proportion of gelator vs substrate. Therefore, it is of utmost importance to study the influence of the drug (to be used for crystallizing in the gel phase) on the gelation process and optimize the conditions for gel phase crystallization.[65]

5.3. Temperature. Most of the reports lack an understanding of the importance of the temperature in the formation of the gel phase. However, the temperature can significantly alter the properties of the gel system if the rate of heating/cooling and the temperature range at which experiments are done fluctuate. The T_{gel} of a system defines the stability of the 3D networks and also controls the morphology of gel fibers. At different temperature points, aggregation of gelators will result in different morphologies (fibrils, nanotubes, tapes, helical, etc.). Weiss and co-workers reported the effect of temperature on the gel properties based on the nature of the 3D network. Gels formed by incubation at 28 °C show thixotropicity, whereas incubation at 30 °C showed no such behaviors.[95] This change in property is due to the formation of fiber-like SAFIN units at 30 °C and spherulitic structure at 28 °C. The cooling rate can also affect the nature of fibers in 3D networks. At high cooling rates, nucleation occurs at a rate faster than that of crystal growth, which results in the formation of a high number of smaller crystals (i.e., fibers and fibrillar spherulites). In contrast, at low cooling rates, the molecules have enough time to diffuse and accommodate on the nucleus surface, and the crystal growth rate is favored over the nucleation rate. Thus, larger crystals (i.e., spherulites) are developed at lower cooling rates. Moreover, when the gel was developed using a higher cooling rate, the gel exhibited better rheological properties. In gel phase crystallization, the heat−cooling rate can affect the crystallization outcomes. This heat−cooling rate may result in a competition between gelation and crystallization, ultimately leading to disruption or strengthening of the gel phase.

6. GEL-CRYSTAL INTERCHANGE

Gelators in the gel phase are a kinetically trapped metastable form, which may spontaneously transform to a more thermodynamically stable crystalline form. Both the gel and crystal depend on the agglomeration and dissociation process of the gelator. The gelators in a solution can adapt to three different situations where they can have (i) a highly ordered aggregation giving rise to crystals, that is, crystallization, (ii) a random aggregation resulting in an amorphous precipitate, and (iii) an aggregation process intermediate between these two extremes resulting in gels. Many a time during crystallization, the whole solution turns into a gel instead of generating

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crystals due to the faster kinetics of fiber growth. Yin et al. reported that in the crystallization process of cefotaxime sodium (CTX) initially formed nanometer/micrometer-sized crystals in solution turn into a gelled phase.[96] Experimental support of nucleation induction time and high interfacial tension imply the strong interaction between crystal surface and solvent molecules, which hindered the arrangement of CTX molecules on the crystal surface to grow (Figure 21). Thus, the slow rate of formation of "primary nuclei" favors the colloidal state, and strong solvent interactions trigger the gel formation of CTX during crystallization.

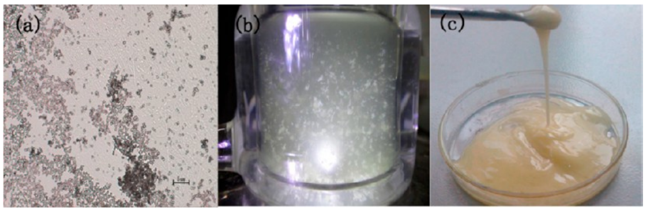

Figure 21. Pictures of different stages CTX crystallization: (a) CTX solution before gelation; (b) gel formed in the crystallizer; (c) gel after the removal of nongelled solvent.[96]

Gao et al. reported a similar observation of getting a jelly like phase (JLP) of cefotaxime sodium during crystallization in polar solvents.[97] They used FT-IR and in situ Raman spectroscopy to demonstrate the role of H-bonding interaction between the CTX and solvent in the formation of JLP. They observed that the JLP is the kinetic form that eventually transforms into crystals and suggested that JLP-crystal transition will make it possible to optimize the crystal form and habit that are difficult to crystallize out in solution.[76] Another example is that during the crystallization of the drug Moxidectin, the addition of water into the solution triggers gelation. Similar to the CTX case, the tiny crystal of Moxidectin and H-bonds between Moxidectin and water molecules favors gel formation.[98] But the metastable gel phase of Moxidectin spontaneously transforms to stable form I. Thus, both the formation of crystal and gel is due to the aggregation and dissociation of the tiny crystal particles. This spontaneous conversion "gel−crystal" is a dynamic process, and there is always a competition between gelation and crystallization (Figure 22).

Since the gel phase is a solvent-selective phenomenon, gelator−solvent interactions are delicate for their stability. Thus, changing the solvent or its properties can impact the gelling ability of the gelator in different solvents during crystallization. Wan and his group reported unusual behavior of 4-(4′-ethoxyphenyl)phenyl-*β*-*O*-D-glucoside (C4BG-1) in solution. In acetonitrile C4BG-1 crystallizes out but forms gel in solvents like acetone, *n*-butanol, 1,2-dichloroethane, and water/1,4-dioxane solvent mixture.[99] Interestingly, gel formed in water/1,4-dioxane collapsed and underwent a gel-crystal transition to give needle-like crystals similar to Moxidectin's transition of gel to form I. Crystals obtained from the collapsed gel and acetonitrile solution had the same diffraction pattern, indicating identical molecular packing. However, they observed that the xerogel had a bilayer structure, where the alkyl part was interdigitated. They suggested that due to strong surface tension between liquid and gel fibrils, fibrils cannot manage the bilayer structure, and the molecular sheets glided into each other where the tails inserted into the interspace of biphenyls and weakened the $\pi-\pi$ interaction, ultimately collapsing the three-dimensional network of the gel phase. This illustrated the role of molecular packing of a gelator on its gelling ability, stability, and transition to the crystalline phase. In the above section, we discussed the polymorphic outcome of Ag-NO_3:PQ5U in different gelling solvents. Gelator's relative ligand conformation (transoid in Forms I−III and cisoid in Form IV) led to different outcomes that were dependent on the compromise between the minimization of the crystal energy and molecular energy. The rate of solvent evaporation played a crucial role in ligand conformations in the Ag(I) complex. Fast evaporation (gel dried in an open vial) showed a 3D network of branched and entangled fibers (Figure 19), whereas a slower rate (dried in the capped vial) allowed the ligand sufficient time to change conformation (energetically favored) to be crystallized out.[93] Another interesting example of phase transformation during crystallization is Valnemulin hydrogen tartrate (used for treating swine dysentery in animals), which undergoes a three-step transformation; first, the amorphous state (A) transforms into a gel (G), which acts as an intermediate phase to give finally a crystalline state (C).[100] They used focused beam reflectance measurements (FBRM) and particle vision measurements (PVM) for in situ monitoring of particle amounts and particle sizes during the transformation of A-G-C in DMF/BuOAc (5% v/v) mixed solvent. During the cooling steps of the crystallization process of Valnemulin hydrogen tartrate, the particle count changes in three steps, first indicating the formation of amorphous material, which suddenly dropped during gel formation and again sharply increased. Samples with a sharp increase in particle count resulted in a crystalline state that was examined using XPRD and DSC. FBRM data and PVM images of valnemulin hydrogen tartrate during the whole crystallization process are shown in Figure 23.

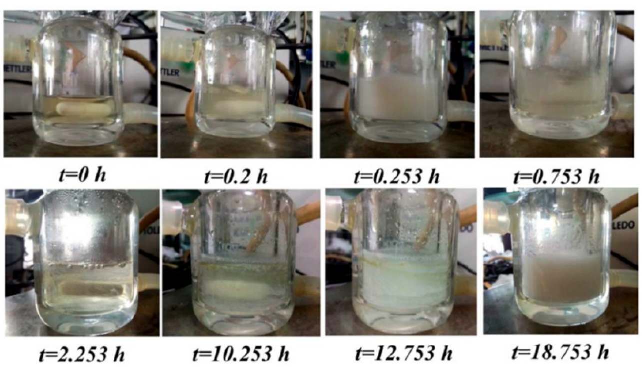

Figure 22. Time scale of sol−gel (gel-crystal) transformation in Moxidectin solution.[98]

The whole transformation process can be divided into two steps: (1) after nucleation, the amorphous state particles were absorbed into the liquid to form a gel structure, and (2) as the temperature decreased, the network structure of gels was destroyed, and the internal structure of the particles was rearranged into the crystalline state. In this case, amorphous particles that appeared first act as gelators and interact with solvent molecules to form a temporary 3D-network structure to give a gel phase. Here the instability of the amorphous phase results in the final crystalline phase, where the gel formation is the intermediate or transitional stage during the whole crystallization process. The development of better imaging

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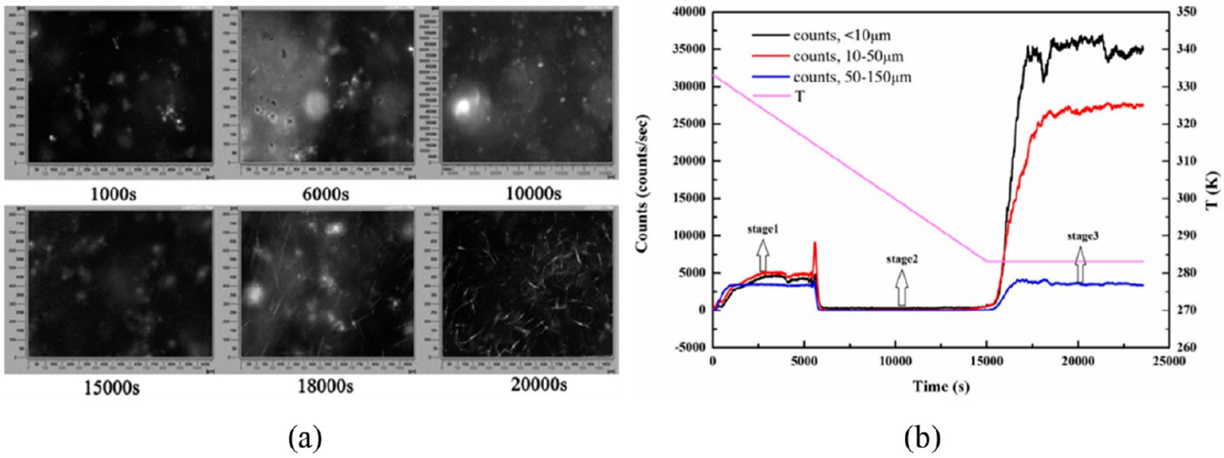

Figure 23. (a) PVM images of valnemulin hydrogen tartrate, (b) change of FBRM data during gel formation and subsequent transformation of Valnemulin hydrogen tartrate.[100]

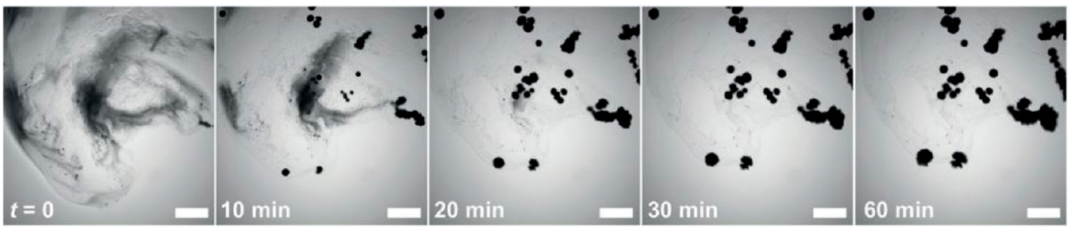

Figure 24. Gel to crystal transition of a pentapeptide gelator under a microscope.[103]

techniques has improved to enhance the study of the kinetics of transition and the complex pathways associated.[101] The mechanism for gel-to-microcrystal transformations for a two-component metastable gel was attributed to the lateral aggregation of nanofibrils into microcrystalline fibers supported by FESEM and NMR analysis.[102] The sporadic transformation of gel to crystal was visualized with a pentapeptide gelator by microscopy and microrheology. The transformation was found to follow a sigmoidal kinetic profile. The phase transformation was not homogeneous, as the less stable gel domains were more susceptible to phase transition. The transition was observed to be facilitated by an elevated temperature and direct physical impairment. However, the absence of water of crystallization inhibited the transformation (Figure 24).[103]

Adams and co-workers very recently demonstrated the marked distinction between the gel phase and the crystalline phase of a functionalized dipeptide gelators (2*S*)-2-(2*S*)-2-{2-naphthalen-2-yloxyacetamido}-3-phenylpropanamido-3-phenylpropanoic acid (2NapAA). The pH-dependent gel to crystal transition was monitored by SAXS and WAXS, which affirms the appearance of crystals once the pH decreases below 4.1. The rate of crystallization and the dimensions of the crystals depend on the rate of the pH change. The pH of the medium was controlled by varying the amount of glucon-δ-lactone (GdL) added to the system. At high concentrations of GdL, the gel fibers are stiffened, and as the pH drops there is an increase in the persistence length along the fiber axis due to reduced flexibility of the gelator molecules induced by charge removal. This example does act as a caution for the tendency to correlate the gel phase by extrapolating the crystal structures of the gelators.[104] Andrews et al. reported the supramolecular gelation behavior of a monoiodinated 2,4,5-triphenyl imidazole derivative in methanol and observed spontaneous crystallization of the gelator, forming a series of distinct polymorphs, one after the other (Figure 25).[105] This gel is in a thermodynamically metastable but kinetically accessible state in the system. Gelation, therefore, occurs first, followed by a stepwise crystallization, forming three increasingly stable methanol solvates.

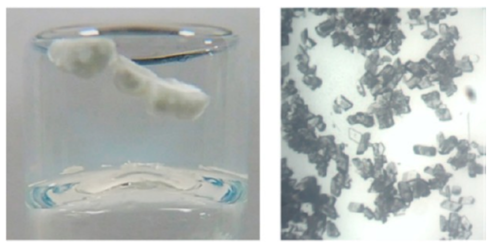

Figure 25. Spontaneous breakdown of the gel to form crystals. Small needle-like crystals initially grown within a 2% w/v I-TPI gel containing 2% w/v DTA (left). Block-like single crystals formed after the breakdown of the same gel (right).[105]

During gel phase crystallization of the substrate molecule, it produces a similar stepwise phenomenon, yielding two increasingly stable solid forms of a drug−gelator salt. Thus, the presence of suitable functional groups, solvent properties, molecular packing arrangement, and steric environment are the balancing factors responsible for the metastable gel state over other stable phases (crystals, precipitate, fibers, etc.). The utility of gel phase crystallization demands a detailed understanding of the media so that reproducibility and selectivity can be achieved at scales greater than what is practiced at the laboratory scale.

7. CONCLUSION

Strategic design and characterization of LMWGs as crystallization media for pharmaceutical compounds are an emerging research expanse in the development of APIs screening and manufacturing processes. However, it is too early to generalize the relationship between the thermodynamic and kinetic parameters of gel systems to design a specific gelator that can gel a given specific solvent that follows a strategic crystallization process. The crystallization events in the gel phase can lead to various observations, essentially the discovery of new solid forms, an increase in selectivity toward a particular solid form, and habit modification of the crystalline outputs. The introduction of specific functional fragments into the gelator molecules to mimic the substrate is a more deliberate approach to engineer the intermolecular interactions such as hydrogen bonding, $\pi-\pi$ interactions, and interactions among the like or complementary functional groups of the gelator and substrate, which is vital. Understanding the role of different variables that influence the molecular packing arrangement in gel and crystal phases is the key to correlate with "gel-crystal" transition, preferential growth of kinetic forms of drugs, breakdown of the gel matrix, and occurrence of gel-drug cocrystals. Understanding gels is therefore completely interdisciplinary in nature, which stands at the crossroad of a wide range of curiosities. A collaborative assembly between inspired researchers will surely lead to enhanced gel applications including one controlling API nucleation. We anticipate that this Perspective will invigorate researchers to take up additional challenges to answer questions surfacing in gel research.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.cgd.3c01211.

> Tabulated examples of some notable applications of LMWGs in various fields (oil spill remediation, dye adsorption, heavy metal purification, sensing) and a list of representative APIs crystallized in gel media (PDF)

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Notes

The authors declare no competing financial interest.

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